

Why Do Workers Dislike Inflation?

Wage Erosion and Conflict Costs

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Why do Workers Dislike Inflation?

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- Prior work: **prices rise faster than wages** [Fischer & Modigliani '78; Shiller '97; Stantcheva '24]
- Relevant: how fast nominal wages **keep up with** prices, i.e., how fast **real wages** recover
[Kessel & Alchian '60, Blanco et al '23]

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[Kessel & Alchian '60, Blanco et al '23]

Instead, this paper:

- Workers must take **costly actions** (“**conflict**”) to have wages catch up with prices
- Substantial welfare costs of inflation **even if** real wages do not fall during inflations

Motivating survey: workers get wage growth though **costly conflict** w/ firms

[E.g. tough conversations with employers; union strike; soliciting offers from other firm]

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“Menu-cost style” (***conflict-cost***) model for wage setting:

[Nakamura–Steinsson '10, Alvarez et al. '16, Auclert et al. '21]

- Workers consider costly conflict w/ firm to raise nominal wage [+ idiosyncratic shocks]

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Analytical results:

1. Impact of inflation on worker welfare determined by **wage erosion**
 - How inflation lowers real wages if conflict decision do not change
2. Wage catch up after inflation due to **costly conflict** doesn't raise workers' welfare
 - On the margin: benefit of wage catch-up are offset by cost of conflict [envelope theorem]

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Takeaway: welfare costs of inflation in labor market even if real wages unaffected by inflation

Quantifying the Costs of Conflict

Key terms for welfare: size of conflict cost and degree of inflation indexation in default wages

Survey evidence to quantify *conflict-cost* model:

1. Conflict threshold = **1.75% of wages**
2. $\approx$ no indexation: 1 p.p. $\uparrow$ inflation $\Rightarrow \sim 0$ p.p. $\uparrow$ nominal wage growth absent conflict

Calibrated model for the **post-Pandemic inflation**:

- **Inflation costs** incorporating conflict $\approx$ **2 times** larger than implied by real wages

Inflation and Conflict. [Lorenzoni & Werning '23 a,b]

- **This paper:** related but distinct notion of conflict—costly actions by workers raises wages
- Our focus is the **welfare costs of inflation**, theirs is the causes of inflation

Costs of inflation. [Baily '56; Friedman '77; Fischer & Modigliani '78; Imrohoroglu '92; Lucas '00; Burstein & Hellwig '08; Alvarez et al '16; Nakamura et al '18; Binetti et al '24]

- **Previous:** inflation costs from range of mechanisms
[e.g. shoe leather costs, menu costs, relative price distortions, inflation volatility, tax distortions]
- **This paper:** new + significant costs of inflation via labor markets

Costs of inflation via labor market. [Shiller '97; Hadjini et al '22; Pilossoph & Ryngaert '23; Auclert '21; Del Canto et al '23; Ferreira et al '23; Pallotti et al '23; Stantcheva '24; Afrouzi et al. '24; Pilossoph et al '24]

- **This paper:** real wage growth misses costs of inflation in labor market due to conflict

Survey Design

Our Survey

- Survey: Prolific
- 3000 prime-age, employed US workers, Feb-March '24
- Representative of the general population in terms of chosen observable characteristics
 - gender, education and political affiliation
- Survey part I [now]: Qualitative evidence to motivate model
 - Workers achieve wage growth through costly conflict with employers
- Survey part II [later]: Quantitative evidence to discipline the model

Representation

Motivating Evidence: Nominal Wage Growth, Inflation and Conflict

Motivating Evidence: Wage Growth, Inflation and Conflict

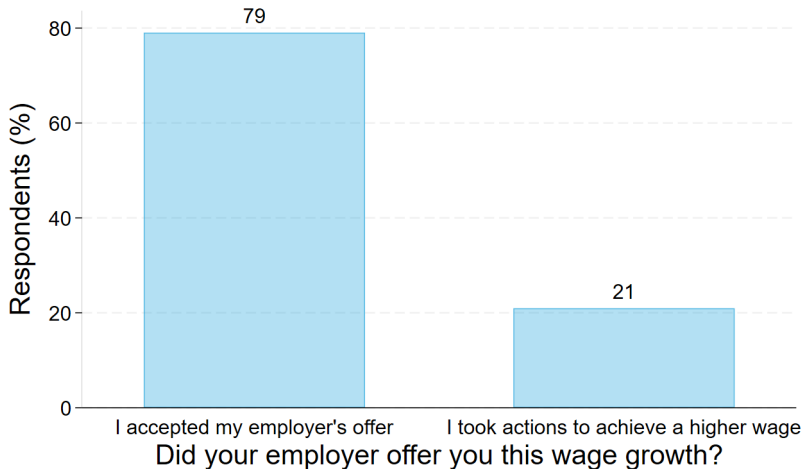
Motivating evidence from survey:

1. Workers **choose** between accepting employer's "default wage" or conflict
2. Conflict **raises wages**
3. Workers who did not conflict believe **conflict could have raised wages**
4. Inflation leads to conflict: 1 p.p. $\uparrow$ inflation $\Rightarrow$ 1.5 p.p. $\uparrow$ prob. of worker conflict
 - 4.1 Conflict motivated to catch-up with inflation

Workers Choose Employer Offer vs. Conflict to Raise Wage

Heterogeneity

By income



Question asks about **nominal** wage growth

Motivating Evidence: Wage Growth, Inflation and Conflict

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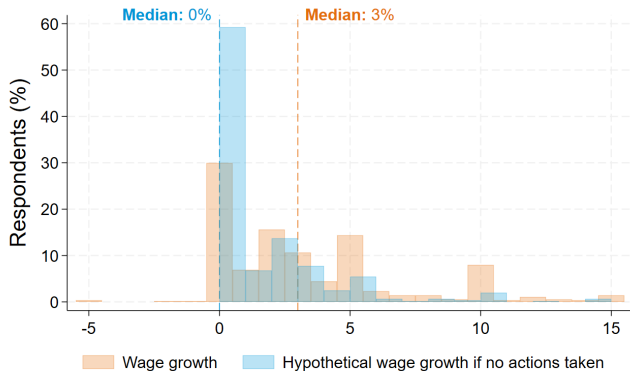
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Conflict Leads to Higher Wages...

Within-individual

High wage workers





- Workers who conflict increased their wage relative to default employer offer.

Motivating Evidence: Wage Growth, Inflation and Conflict

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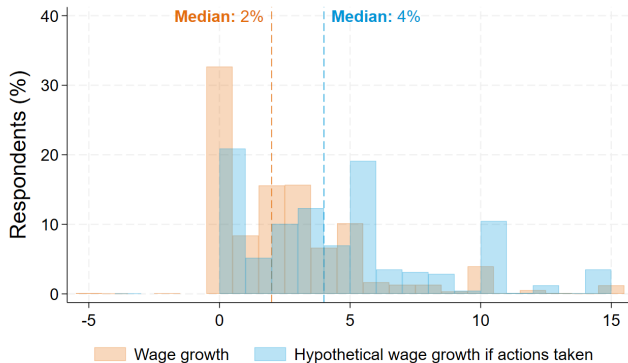
Workers Sacrifice Wage Growth To Avoid Conflict

Within-individual



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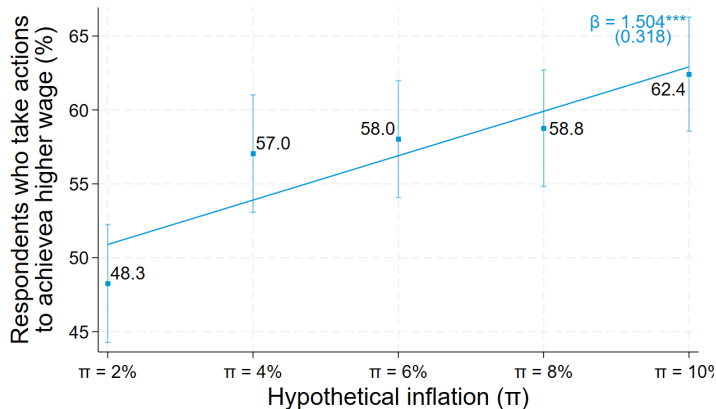
- Workers who do not conflict, report having sacrificed 2pp of wage growth

Motivating Evidence: Wage Growth, Inflation and Conflict

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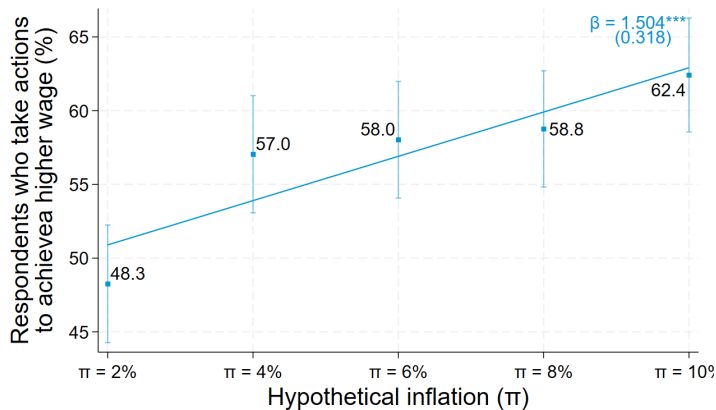
Inflation Increases the Probability of Conflict



Q: consider a hypothetical situation, inflation is expected to be $x\%$ in next 12 months.

- Where $x \in \{2, 4, 6, 8, 10\}$

Inflation Increases the Probability of Conflict

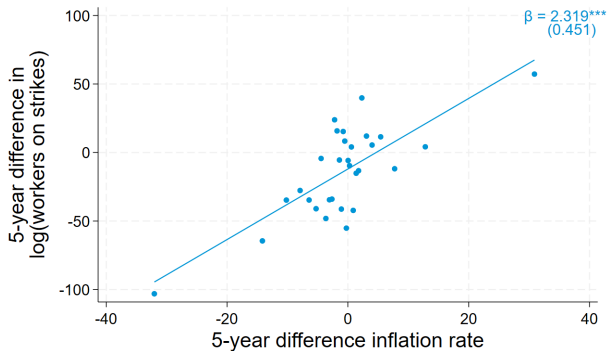


- 1% $\Delta\pi$ implies a 1.5% increase in conflict – **state-dependent conflict decisions**

Inflation is associated with higher union activity

No fixed-effects

Specification

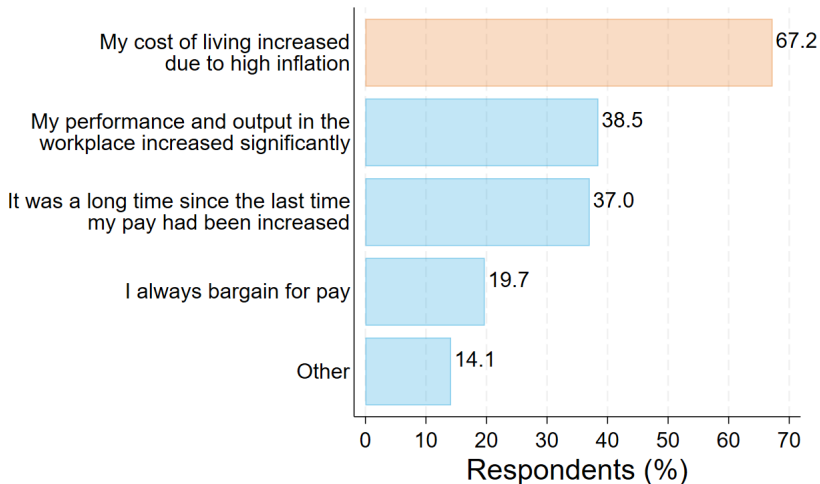


$$\Delta \log 100 \times (\text{workers involved in strikes})_{i,t,t-5} = \gamma_i + \lambda_t + \beta \Delta \pi_{i,t,t-5} + \varepsilon_{it},$$

International Labor Organization data, 1960 – 2020

... And Catch-Up with Inflation

Motivations for conflict:



Motivating Evidence: Wage Growth, Inflation and Conflict

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Next: incorporate these features in a "menu-cost style" model of wage setting

[Alvarez et al '16; Nakamura et al. '18; Auclert et al '24]

Analytical Results:
Costs of Inflation from Wage Erosion and Conflict

Workers' Problem

- A continuum of workers $i \in [0, 1]$ balances **benefit of wage catch up** and **cost of conflict**

$$\max_{\{\mathcal{I}_{i,t}\}_{t=0}^{+\infty}} \mathbb{E}_0 \left[\sum_{t \geq 0} \beta^t (\log c_{i,t} - \kappa_{i,t} \mathcal{I}_{i,t}) \right],$$

where $\mathcal{I}_{i,t} = 1$ if the worker takes costly actions to increase pay and 0 otherwise.

- Now: hand-to-mouth $c_{i,t} = w_{i,t}$. [Paper: borrowing constraints.]
- $\kappa_{i,t}$: i.i.d. “Calvo-plus” cost to increase pay [Nakamura & Steinsson '10; Auclert et al. '23]

$$\kappa_{i,t} = \begin{cases} \kappa & \text{with probability } 1 - \lambda \\ 0 & \text{with probability } \lambda \end{cases}$$

Reduced form cost captures a variety of conflict actions

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where $\mathcal{J}_{i,t} = 1$ if the worker takes costly actions to increase pay and 0 otherwise.

- Real wage (α : default nominal wage growth without inflation & γ : degree of indexation)

$$w_{i,t} = \begin{cases} w_{i,t-1} e^{\alpha - (1-\gamma)\pi_t} & \text{if } \mathcal{J}_{i,t} = 0 \\ w_{i,t}^* & \text{if } \mathcal{J}_{i,t} = 1 \end{cases}$$

Indexation $\rightarrow$ some wage changes not associated with conflict

- Conflict-induced (real) wage $w_{i,t}^*$, keeps up w/ inflation & productivity [endogenous in GE]

$$\log w_{i,t}^* = \log w_{i,t-1}^* + g_z + z_{i,t},$$

$z_{i,t} \in [-\bar{z}, \infty)$ is idiosyncratic productivity shock, g_z is trend productivity growth.

The Impact of Inflation on Worker Welfare and Wages

- An unexpected $t = 0$ shock to inflation, $\{\hat{\pi}_t\}_{t=0}^{+\infty} = \{\pi_t - \pi^{ss}\}_{t=0}^{+\infty}$. Perfect foresight after.
- Study its impact on aggregate worker welfare

$$\mathcal{W} \equiv \int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \{ u(c_{i,t}) - \kappa_{i,t} \mathcal{J}_{i,t} \} \right] di$$

and aggregate log real wages

$$\log w_t \equiv \int_0^1 \log(w_{i,t}) di.$$

Model: Higher inflation increases share of conflict

Proposition.

*If $\gamma < 1$, then an **increase in inflation** at $t = 0$ leads to a **larger fraction** of workers engaging in **conflict** at $t = 0$.*

- State dependency: conflict likelihood increases with inflation
- Unless default wages were fully indexed ($\gamma = 1$)

The Impact of Inflation on Worker Wages

Definition 1.

The impact of inflation on aggregate log real wage can be decomposed into

$$\hat{w}_t \equiv \log w_t - \log w^{ss} = \hat{w}_t^{\text{erosion}} + \hat{w}_t^{\text{catchup}},$$

- **Wage erosion**: impact of inflation if worker's conflict decision does not change

$$\hat{w}_t^{\text{erosion}} \equiv \omega_t \left(\pi_t, \mathcal{I}_{i,t}^{ss} \right) - \omega_t \left(\pi^{ss}, \mathcal{I}_{i,t}^{ss} \right),$$

where $\omega_t \left(\pi_t, \mathcal{I}_{i,t} \right)$ is aggregate log real wage at time t given $\pi_t, \mathcal{I}_{i,t}$.

- **Wage catch up**: impact of inflation through changes in conflict decision

$$\hat{w}_t^{\text{catch up}} \equiv \omega_t \left(\pi_t, \mathcal{I}_{i,\tau} \right) - \omega_t \left(\pi_t, \mathcal{I}_{i,t}^{ss} \right).$$

Theorem.

The (first-order) impact of inflation on worker welfare is given by *wage erosion*

$$\underbrace{\hat{\mathcal{W}}}_{\text{impact on worker welfare}} \approx \sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{erosion}} = \underbrace{\sum_{t=0}^{\infty} \beta^t \hat{w}_t}_{\text{impact on aggregate wage}} - \sum_{t=0}^{+\infty} \beta^t \hat{w}_t^{\text{catch up}}.$$

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Because the benefits of *wage catch up* are offset by *costs of conflict*

$$\sum_{t=0}^{\infty} \beta^t \hat{w}_t^{\text{catch up}} \approx \underbrace{\int \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \kappa_{i,t} (\mathcal{I}_{i,t} - \mathcal{I}_{i,t}^{ss}) \right] di}_{\text{impact on conflict costs}}.$$

[Milgrom-Segal '02 envelope theorem applied to discrete choice set]

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- The impact of inflation on *aggregate wage* $\sum_{t=0}^{\infty} \beta^t \hat{w}_t$ *does not measure* welfare
 - Even ≈ 0 , inflation could still be costly because wage catch up comes from costly conflicts

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- Contrast w/ purely time-dependent wage setting: $\kappa \rightarrow \infty$, $d\mathcal{W} = \sum_{t=0}^{\infty} \beta^t \hat{w}_t$
 - Special case in which aggregate real wage sufficient for welfare

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 - Special case in which aggregate real wage sufficient for welfare
- Conflict cost are *aggregate* deadweight loss, not just redistribution from workers to firms
- Extensions: (1) other aggregate shocks, (2) beyond hand-to-mouth, (3) unemployment, (4) general random menu costs

Measuring Costs of Conflict with Survey

Additional Survey Evidence to Quantify and Validate the Model

Key terms for welfare: size of conflict cost and degree of inflation indexation in default wages

Calibrate the model and quantify the cost of conflict

- **Elicit the cost of conflict**
- Elicit how default wage offer changes after inflation (absent conflict)

Eliciting Costs of Conflict

- Elicit nominal wage growth workers believe they get **via costly actions**

$$\Delta W^{\text{conflict}} \equiv \log W_{i,t}^* - \log W_{i,t-1}$$

[$W_{i,t}^*$: conflict-induced nominal wage]

What pay growth in next 12 months do you think you would get if you do your best to increase pay using any strategies at your disposal, including the common strategies listed above?

Eliciting Costs of Conflict

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[$W_{i,t}^*$: conflict-induced nominal wage]

What pay growth in next 12 months do you think you would get if you do your best to increase pay using any strategies at your disposal, including the common strategies listed above?

- Elicit nominal wage growth where workers are **indifferent** between
 - accepting** if offered by their employers vs **taking costly actions** ΔW^{indiff}
- Cost of conflict:

$$x^{\text{conflict}} \equiv \Delta W^{\text{conflict}} - \Delta W^{\text{indiff}}$$

Eliciting Costs of Conflict

Standard tool from experimental economics: “multiple price list”

[Jack et al. '22]

	I would accept my employer's pay growth offer	I would do my best using any strategies at my disposal to increase my pay further
Employer offers you pay growth of 4%	☐	☐
Employer offers you pay growth of 3.5%	☐	☐
Employer offers you pay growth of 3%	☐	☐
Employer offers you pay growth of 2.5%	☐	☐
Employer offers you pay growth of 2%	☐	☐
Employer offers you pay growth of 1.5%	☐	☐
Employer offers you pay growth of 1%	☐	☐
Employer offers you pay growth of 0.5%	☐	☐
Employer offers you pay growth of 0%	☐	☐

Consider respondent who reports $\Delta W^{\text{conflict}} = 4\%$, then get list of potential offers $\leq 4\%$

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Employer offers you pay growth of 0%	☒	☐

Respondent never engages in conflict

If they report they would always accept offer → never conflict – high conflict cost

Eliciting Costs of Conflict

Standard tool from experimental economics: “multiple price list”

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	I would accept my employer's pay growth offer		I would do my best using any strategies at my disposal to increase my pay further
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If they report they would always take actions → always conflict – zero conflict cost

Eliciting Costs of Conflict

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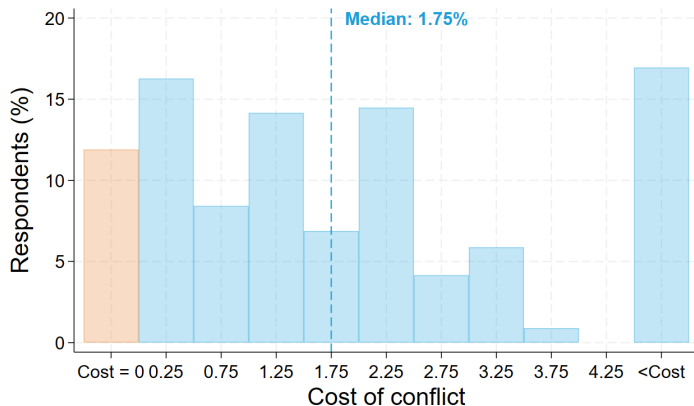
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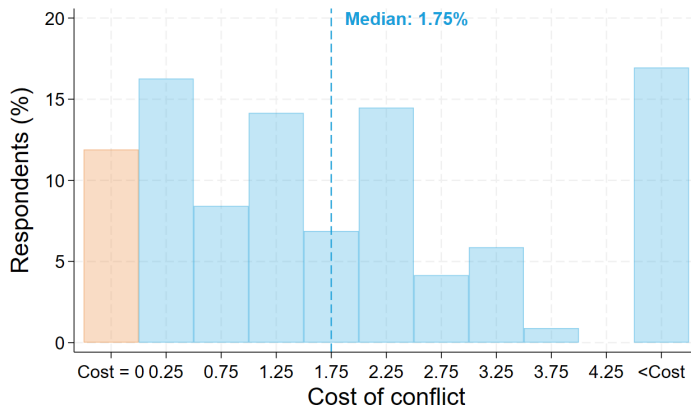
Respondent indifference wage is between 2% and 2.5% pay growth

Let ΔW^{indiff} denote lowest nominal wage growth where workers accept employers' offer, then cost is:

$$x^{\text{conflict}} \in [\Delta W^{\text{conflict}} - \Delta W^{\text{indiff}}, \Delta W^{\text{conflict}} - \Delta W^{\text{indiff}} + 0.5]$$



- Direct evidence on **conflict costs**: $\kappa_{i,t} = 0$ with prob. λ and $\kappa_{i,t} = \kappa$ otherwise
 - $\lambda = 0.11$ to match share of people who would always conflict
 - Calibrate κ to median $x^{\text{conflict}} = 1.75\%$



- In paper, **validation exercises**:

1. **Lower conflict costs** predict **higher likelihood** of having taken **actions** in 2023
2. **Lower conflict costs** predict **less wage sacrificed** from not taking actions in 2023

[Here](#)

[Here](#)

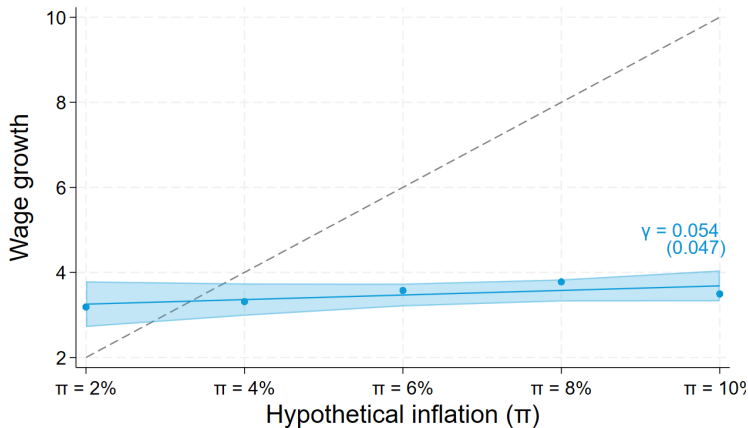
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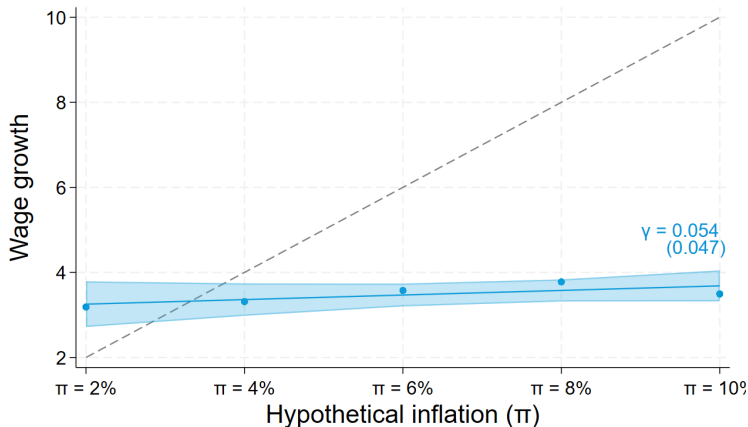
Indexation: Perceived Nominal Wage Growth without Conflict



Question:

- Consider a hypothetical situation, inflation is expected to be $x\%$ in the next 12 months.
- Where $x \in \{2, 4, 6, 8, 10\}$

Indexation: Perceived Nominal Wage Growth without Conflict



- Evidence on **default wages + indexation**: $W_{i,t}^{default} = W_{i,t-1} e^{\alpha + \gamma \pi_t}$
- Indexation: $\Delta \pi = 1\%$ then $\Delta \log W_{i,t}^{default} = 0.05\% \Rightarrow \gamma = 0.05$
 - Intercept: with zero inflation, 3% default wage growth $\Rightarrow \alpha = 0.75\%$

1. **Significant cost of conflict** $x^{\text{conflict}} = 1.75\%$ of wages and $\lambda = 0.11$
2. **Almost no indexation**: 1 p.p. inflation $\Rightarrow \sim 0$ in employer-offered nominal wage growth
 - 2.1 Significant amount of (unrelated) default wage changes

Quantifying the Costs of Inflation: The Importance of Conflict

Calibration - Quarterly Frequency

<i>Parameter</i>	<i>Description</i>	<i>Value</i>	<i>Target</i>
β	Discount factor	0.99	Standard
κ	Conflict cost	7.89%	Survey: such that $\underline{x}^{ss} = -1.75\%$
λ	Share of free adjusters	2.9%	Survey: 11% annual free adj.
g_z	Trend productivity growth	0.50%	2% annual growth
α	Default nom. wage growth	0.75%	Survey: 3% annual def. w. growth
γ	Indexation of default nominal wage	0.05	Survey: indexation
π^{ss}	Steady state inflation	0.5%	2% annual inflation
$z_{i,t}$	Idios. shocks $z_{i,t} + \mu \sim \text{Gamma}(a, b)$	(0.75, 0.003)	$\mathbb{E}[z_{i,t}] = 0$ Survey: 48% annual share of conflict

Three Inflationary Scenarios

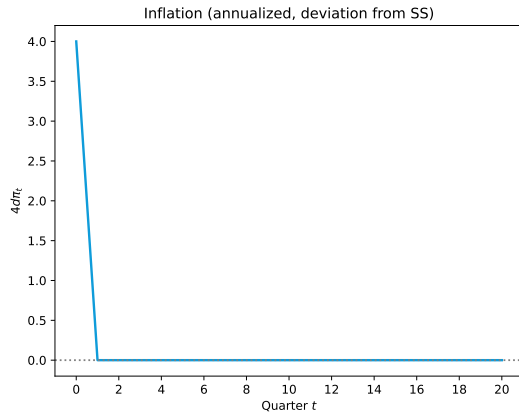
Solve for first-order responses to inflation shock $\{d\pi_t\}_{t \geq 0}$, starting from steady state

- Using SSJ methods [Auclert-Bardóczy-Rognlie-Straub '21, Auclert-Rigato-Rognlie-Straub '24]

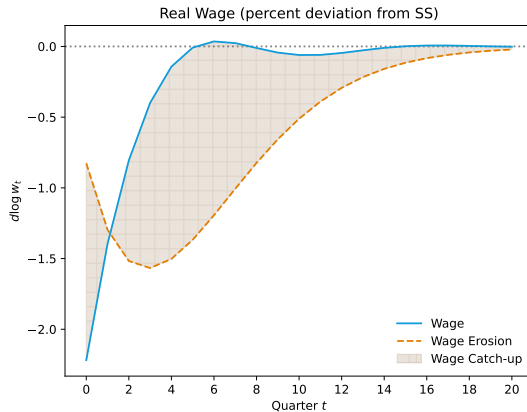
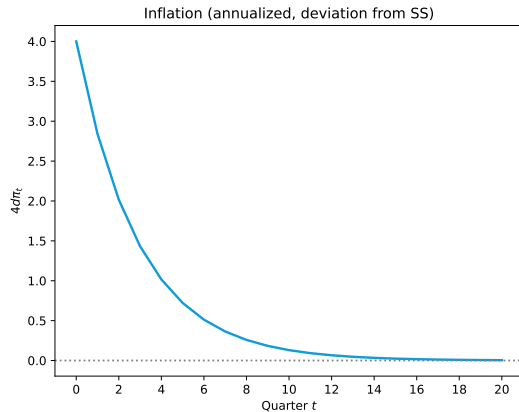
Evaluate three inflation shocks:

1. Transitory inflation: $d\pi_0 > 0$ and $d\pi_t = 0$ for $t \geq 1$
2. Persistent inflation: $d\pi_t = \rho^t \varepsilon$, with $\rho = 0.71$
3. 2021-23 inflation experience

Wage Dynamics: Transitory Inflation



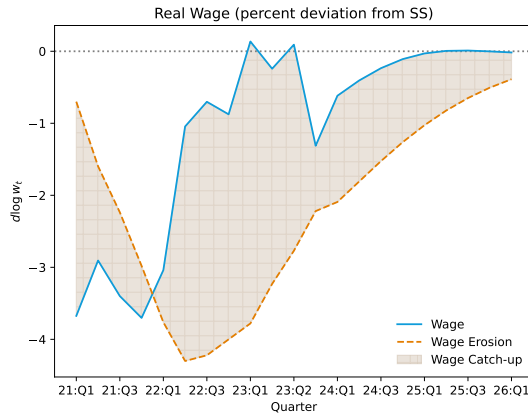
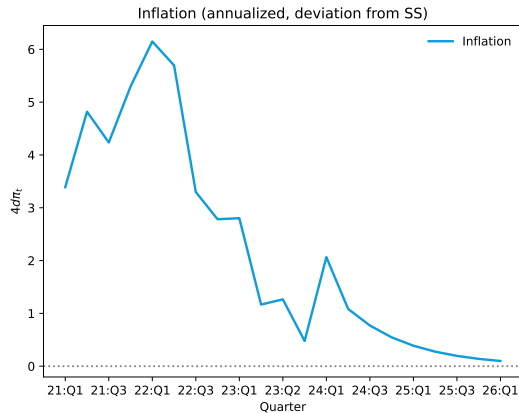
Wage Dynamics: Persistent Inflation



Wage Dynamics: 2021-23 Inflation

Robustness to γ

Robustness to λ



Welfare Decomposition

	<i>Overall Welfare</i>	<i>Real Wage</i>	<i>Agg. Costs of Inflation from Conflict</i>
Transitory inflation	−0.98%	−0.06%	−0.92%
Persistent inflation	−3.30%	−1.29%	−2.01%
2021-2023 inflation	−10.98%	−5.42%	−5.56%

Note: Welfare units = % of 1-year consumption

- For post-pandemic inflation, **welfare costs** ~**twice as large** as inferred from wage response.

Alternative Inflation Expectations

Previous analysis assumed **perfect foresight** of the inflation path

- Did people at the time have the correct expectations for the future path of inflation?

Alternative Inflation Expectations

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SSJ methods can be easily adapted to general expectations [Auclert-Rognlie-Straub '20]

- We now use extend the analysis to alternative models of expectations.

As Bardóczy-Guerreiro (2020), specify **worker's expectations directly** using **survey data**

[Benefit: Do not have to take a stance on wilderness of bounded rationality]

- Expectations from **Michigan Survey of Consumers**

Alternative Inflation Expectations

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[Auclert-Rognlie-Straub '20]

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- Expectations from **Michigan Survey of Consumers**

Also, easy to implement other models of expectations:

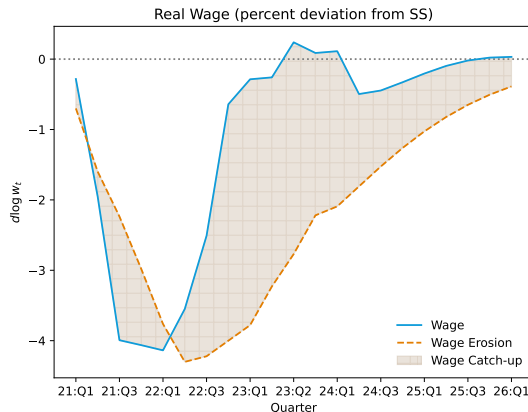
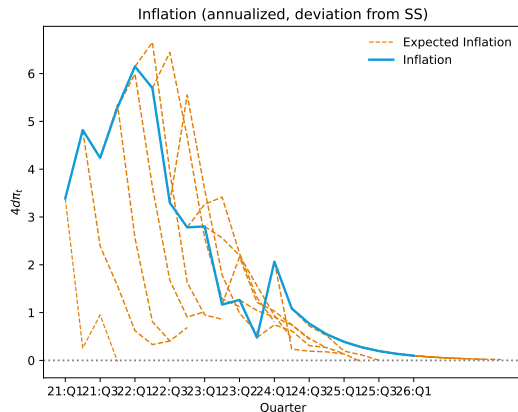
- No foresight [$\mathbb{E}_t[d\pi_{t+h}] = 0$]
- Sticky information [Mankiw-Reis, 2002]

No Foresight

Sticky Expectations

Robustness to inattention parameter

Alternative Inflation Expectations



- With **observed expectations** $\Rightarrow$ Agg. Costs of Conflict mostly unchanged
 - 50% of overall welfare loss $\Rightarrow$ welfare costs double accounting for conflict

Work in Progress/In the Paper

- **Match real wage** dynamics
- **Supply**-driven inflation shocks vs **demand**-driven inflation shocks
- **General equilibrium** analysis
 - Aggregate welfare
 - Quantitative lesson robust to movements in employment
 - Endogeneize conflict-induced wage
- **Beyond hand-to-mouth**
 - A version of envelop theorem still holds

Beyond HtM

Conclusion

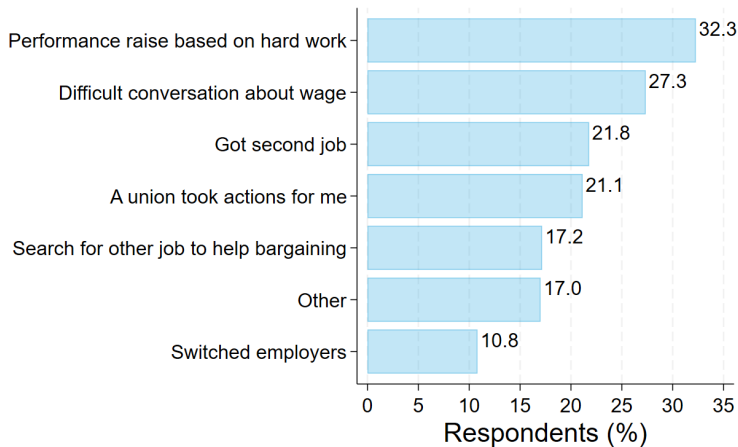
Conclusion

- Basic observation: workers get wage growth though **costly conflict** w/ firms
- Higher inflation raises likelihood of conflict between workers and employers
- Survey shows that workers willing to sacrifice 1.75% of wages to avoid costly conflicts
- Welfare costs of inflation in labor market even if real wages are relatively unaffected during inflations
 - Costs of inflation to workers ≈ 2 **times** larger than that implied by path of real wages

Outline

Extra Slides

What are the Costly Actions?



Takeway: Workers engage in a **diverse set of costly actions**

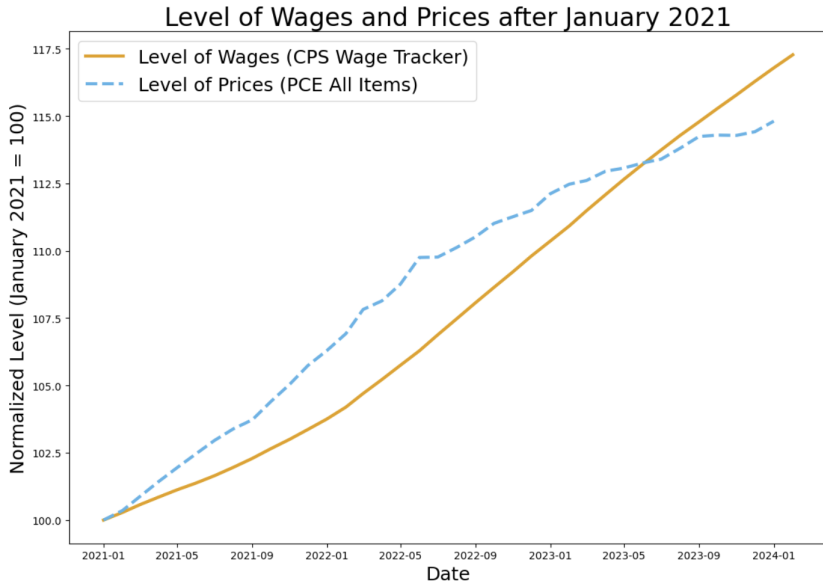


Table 3: Distributions in Survey Sample vs. Population

	Survey	US population
Male	0.52	0.52
Female	0.48	0.48
Secondary education (e.g. GED/GCSE)	0.02	0.02
High school diploma/A-levels	0.37	0.39
Technical/community college	0.12	0.11
Undergraduate degree (BA/BSc/other)	0.32	0.30
Graduate degree (MA/MSc/MPhil/other)	0.14	0.13
Doctorate degree (PhD/other)	0.04	0.04
Democrat	0.28	0.28
Republican	0.26	0.26
Independent	0.33	0.33
None	0.07	0.07
Other party	0.06	0.06
22-29 years old	0.24	0.20
30-39 years old	0.38	0.29
40-49 years old	0.21	0.26
50-60 years old	0.16	0.26

Table 3: Distributions in Survey Sample vs. Population

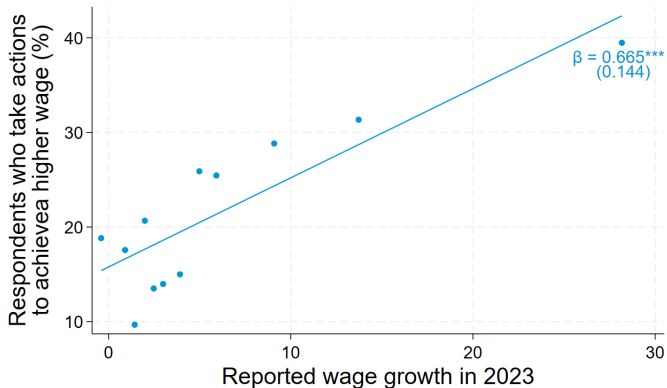
	Survey	US population
Full-Time	0.83	0.83
Part-Time	0.17	0.17
For-profit company	0.80	0.77
Not-for-profit corporation	0.09	0.07
State government	0.03	0.06
Federal government	0.02	0.03
Local government	0.04	0.07
Other employer	0.01	
White	0.68	0.75
Black	0.12	0.14
Asian	0.08	0.07
Mixed	0.08	0.02
Other	0.04	0.02
Not reported	0.00	

Table 3: Distributions in Survey Sample vs. Population

	Survey	US population
Covered by a union	0.11	0.13
Not part of a union	0.81	0.87
Not reported	0.08	
<u>Income</u>		
\$0-\$19,999	0.12	0.12
\$20,000-\$39,999	0.24	0.22
\$40,000-\$69,999	0.34	0.31
\$70,000-\$99,999	0.17	0.16
\$100,000-\$124,999	0.06	0.08
\$125,000+	0.07	0.11

Workers with higher wage growth were more likely to have taken costly actions

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Lemma.

Let the value function be

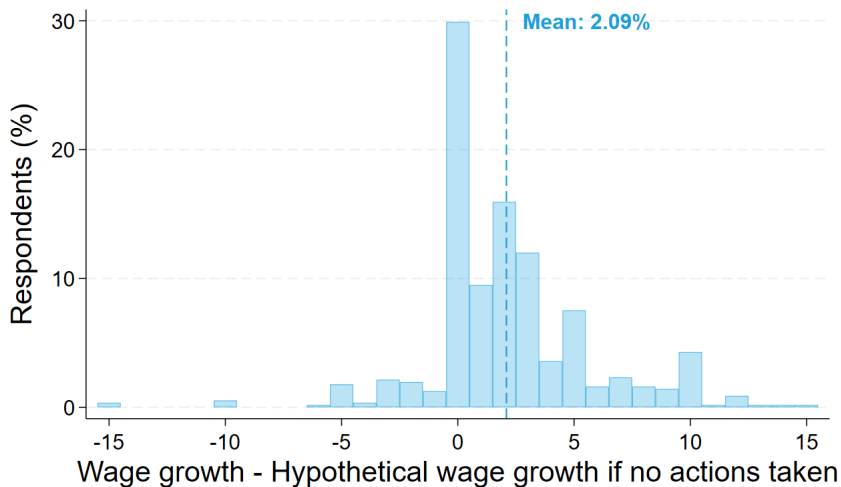
$$v(x) \equiv \max \left\{ x + \beta \mathbb{E} \left[\max \left\{ v(0) - \kappa', v(x') \right\} \right] \right\}$$

where $x \equiv \log(w_{i,t}/w_{i,t}^)$ denotes the wage gap. Then,*

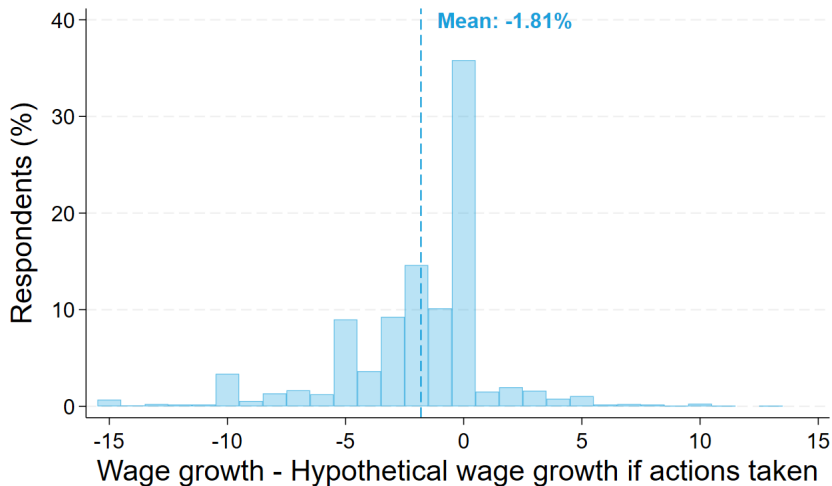
$$v(-x^{\text{conflict}}) = v(0) - \kappa.$$

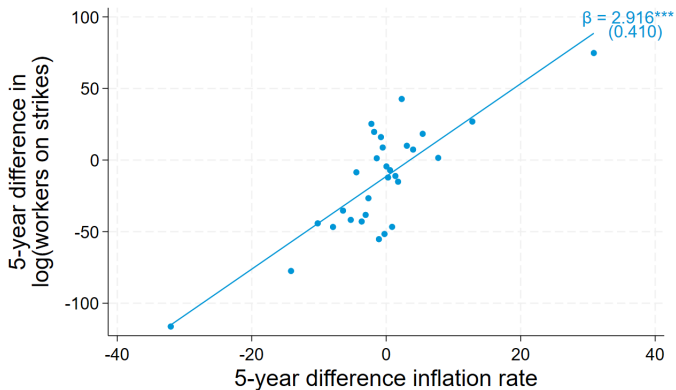
- Measured conflict cost is directly informative of κ .

Conflict Leads to Higher Wages: Within-individual

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Workers sacrifice wage growth not to conflict: Within-individual

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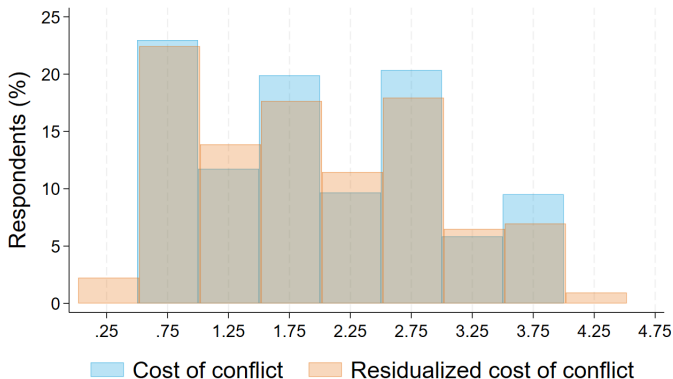
$$\Delta \log(\text{workers involved in strikes}) * 100_{i,t,t-5} = \beta \Delta \pi_{i,t,t-5} + \varepsilon_{it}^1$$

¹Note: Standard errors are clustered at the country level.

$$\Delta \log(\text{workers involved in strikes}) * 100_{i,t,t-5} = \gamma_i + \lambda_t + \beta \Delta \pi_{i,t,t-5} + \varepsilon_{it}$$

- The dependent variable is the 5-year long log difference of "Workers involved in strikes and lockouts", sourced from the International Labour Organization, multiplied by 100 for ease of interpretation.
- As independent variable, we employ the 5-year long difference of headline inflation, sourced by the World Bank, trimmed at 2.5% on each tail.
- Country and year fixed effects are included.
- Observations are unweights.
- Standard errors are clustered at the country level only.
- The data availability per year depends on the country. The analysis includes 78 countries, between 1969 to 2022: Algeria, Australia, Austria, Bangladesh, Barbados, Belgium, Botswana, Brazil, Burundi, Cameroon, Canada, Chile, Colombia, Costa Rica, Cyprus, Denmark, Ecuador, Egypt, Arab Rep., El Salvador, Estonia, Fiji, Finland, France, Germany, Ghana, Greece, Guatemala, Haiti, Honduras, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy, Jamaica, Japan, Kenya, Korea, Rep., Latvia, Lithuania, Malaysia, Malta, Mauritius, Mexico, Morocco, Myanmar, Netherlands, New Zealand, Nigeria, Norway, Pakistan, Panama, Papua New Guinea, Peru, Philippines, Poland, Portugal, Romania, Russian Federation, San Marino, Slovak Republic, South Africa, Spain, Sri Lanka, Suriname, Sweden, Switzerland, Thailand, Trinidad and Tobago, Tunisia, Turkiye, Uganda, Ukraine, United Kingdom, United States, and Uruguay.

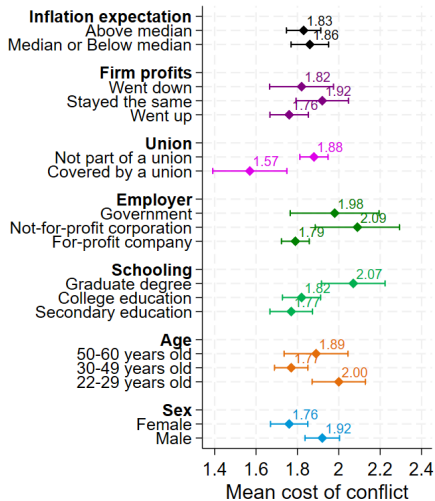
Conflict Costs vs Residualized Conflict Costs

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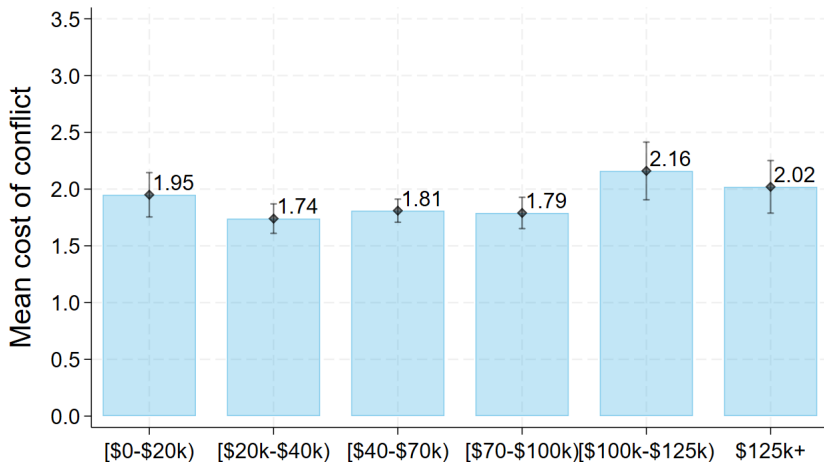
2

²Note: Respondents who never and always engage in conflict have been excluded from the figure and the calculation of the residualized cost of conflict. The residualized cost of conflict has been generated by regressing cost of conflict on dummy variables for the categories of age, education, income and union coverage. The categories excluded were 30-49 years old, income of [100k-125k), graduate education and non-union coverage.

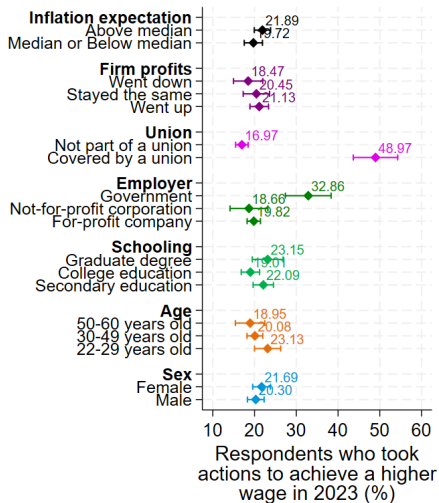
Heterogeneity: Conflict Costs

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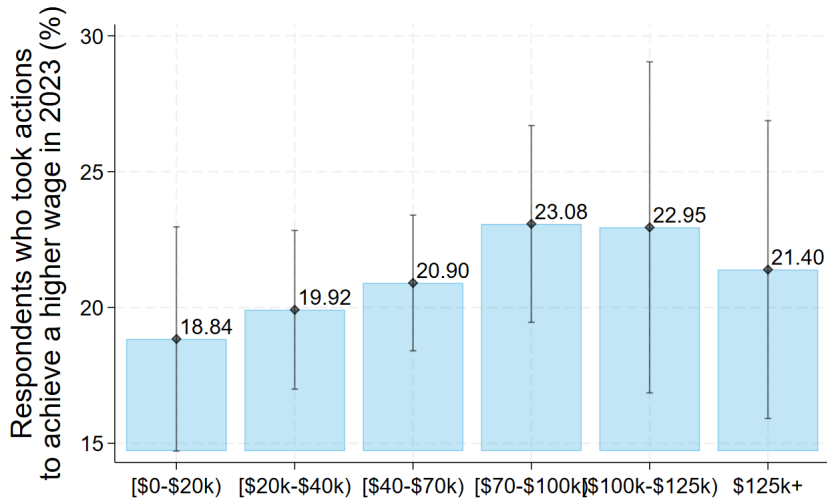
Heterogeneity: Conflict Costs, by income

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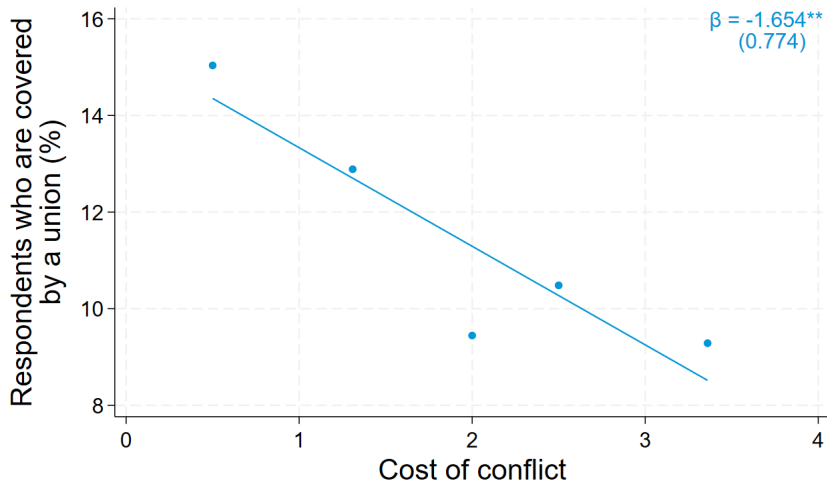
Heterogeneity: Respondents engaging in conflict in 2023

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Heterogeneity: Respondents engaging in conflict in 2023, by income

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Union members have lower cost of conflict

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What if workers can save, subject to standard borrowing constraint $a_{i,t} \geq \underline{a}$?

Theorem.

The (first-order) impact of inflation on worker welfare is given by *corrected-wage erosion*

$$\hat{\mathcal{W}} \approx \sum_{t=0}^{\infty} \beta^t u'(c_{i,t}^{ss}) w_{i,t}^{ss} \hat{w}_{i,t}^{erosion}$$
$$\Rightarrow \frac{\hat{\mathcal{W}}}{\int_0^1 u'(c_{i,t}^{ss}) w_{i,t}^{ss} di} \approx \sum_{t=0}^{+\infty} \beta^t \hat{w}_t^{erosion} + \sum_{t=0}^{\infty} \beta^t \text{COV} \left(\frac{u'(c_{i,t}^{ss}) w_{i,t}^{ss}}{\int_0^1 u'(c_{i,t}^{ss}) w_{i,t}^{ss} di}, \hat{w}_{i,t}^{erosion} \right)$$

- Additional correction: for welfare, not all wage erosion weighted equally
 - Erosion of *borrowing-constrained workers* has higher welfare impact

- Let X be an **arbitrary** non-empty set and $f : X \times [0,1] \rightarrow \mathbb{R}$ be a real-valued function.
- Let $V(\theta)$ be the optimal value as a function of θ and let $X^*(\theta)$ be the set of maximizers.

$$V(\theta) := \sup_{x \in X} f(x, \theta), \forall \theta \in [0,1] \quad \text{and} \quad X^*(\theta) := \{x \in X | f(x, \theta) = V(\theta)\}$$

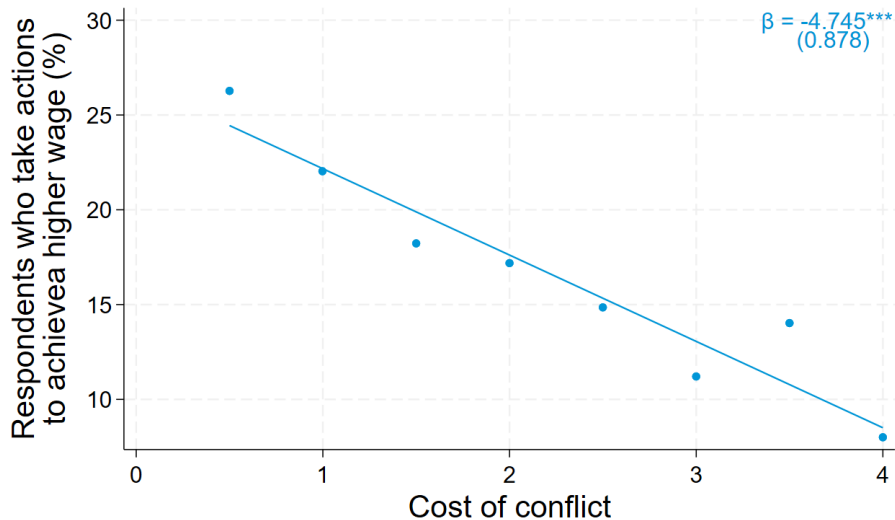
Theorem.

If $f(x, \cdot)$ is differentiable in $\theta \in [0,1]$ for all $x \in X$ and $X^(\theta) \neq \emptyset$ for all $\theta \in [0,1]$, then for any selection of x^* of X^* ,*

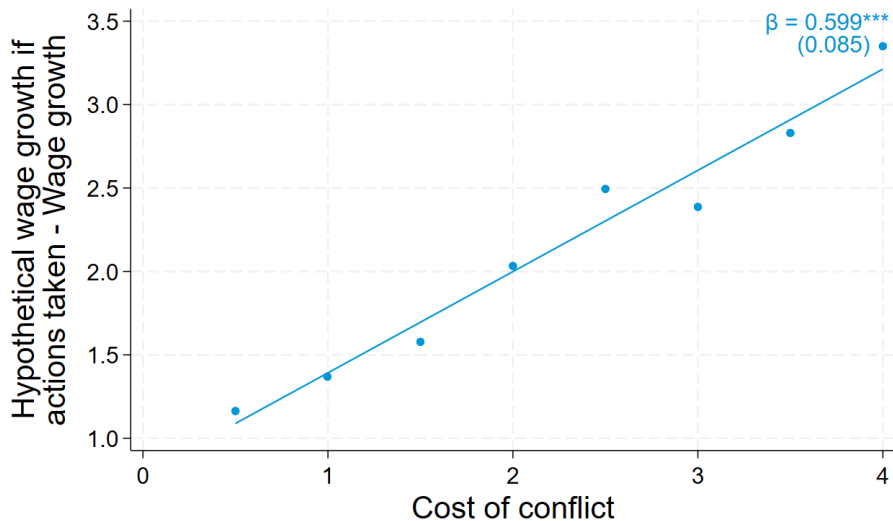
$$V(\theta) = V(0) + \int_0^\theta \frac{\partial f(x^*(z), z)}{\partial \theta} dz \quad \forall \theta \in [0,1] \quad \text{and} \quad V'(\theta) = \frac{\partial f(x^*(\theta), \theta)}{\partial \theta} \quad \text{a.s.}$$

- For us, for each worker i , $\theta \leftrightarrow \{d\pi_t\}_{t=0}^{+\infty}$ & $x \leftrightarrow \{\mathcal{J}_{i,t}\}_{t=0}^{+\infty}$ (conflict choice, $\mathcal{J}_{i,t} \in \{0,1\}$)
- The envelope theorem holds a.s. at the worker level, and everywhere at the aggregate level

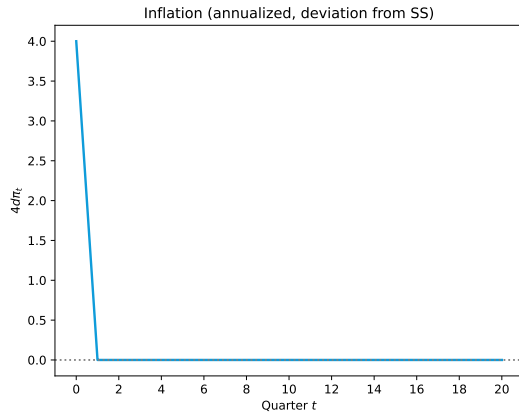
Lower Conflict Costs Predict Costly Actions

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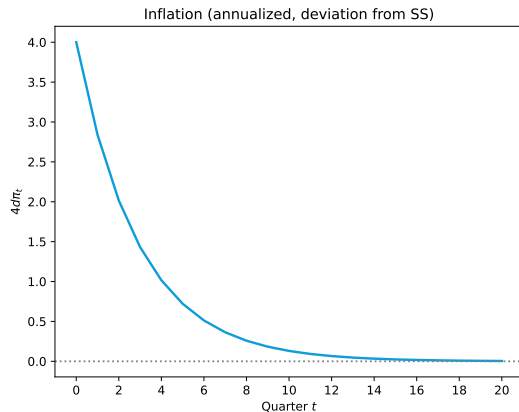
Lower Conflict Costs Predict Lower Sacrifice from Not Taking Actions

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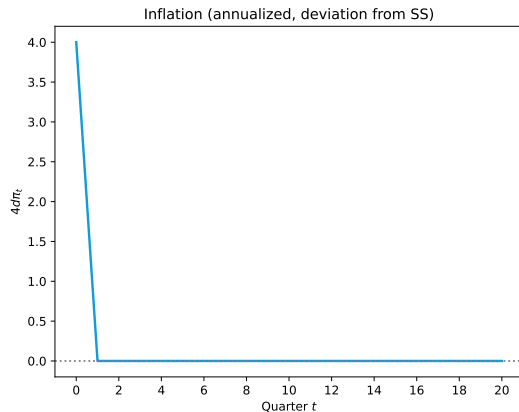
Wage Dynamics: One-period Inflation



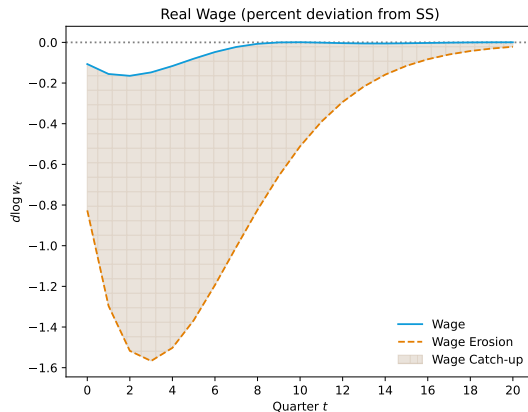
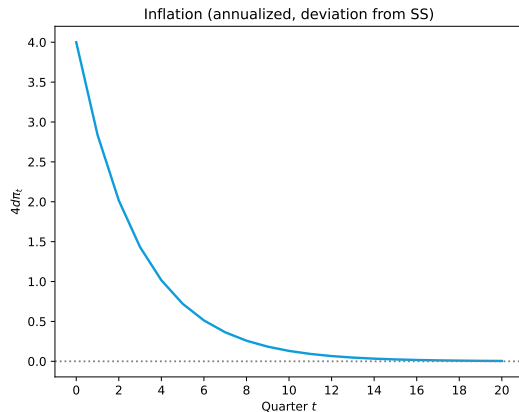
Wage Dynamics: Persistent Inflation

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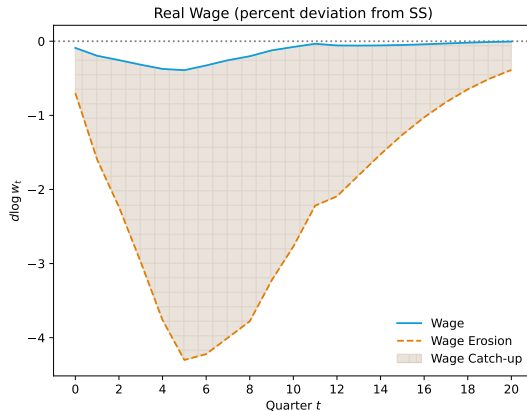
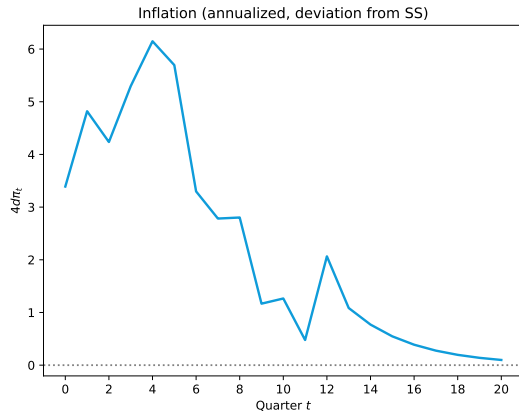
Wage Dynamics: Transitory Inflation without foresight

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Wage Dynamics: Persistent Inflation without foresight

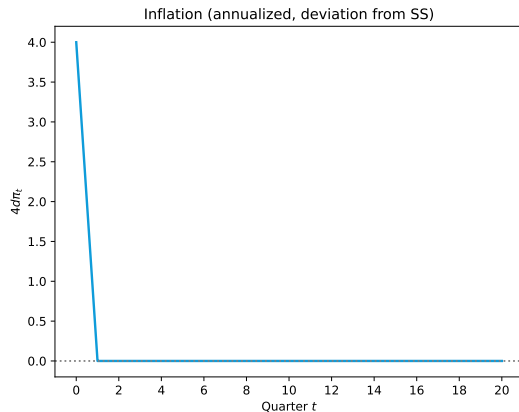
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Wage Dynamics: 2021-23 Inflation without foresight

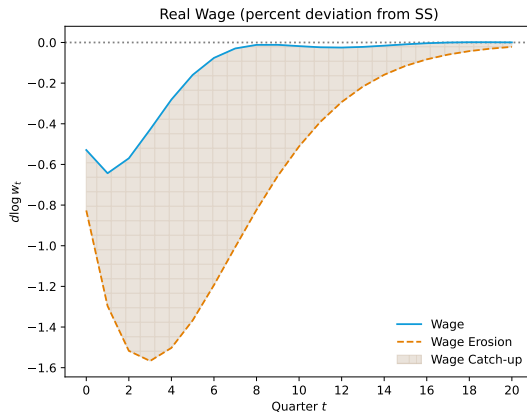
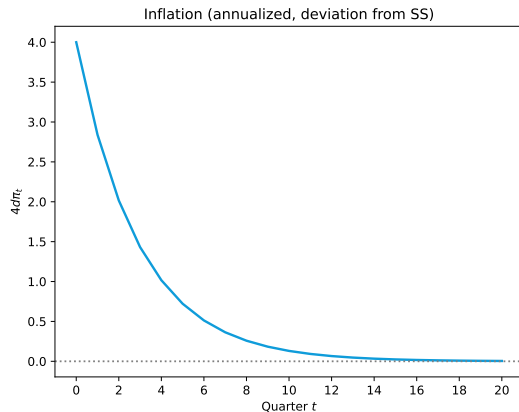
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	<i>Overall</i>	<i>Wage response</i>	<i>Conflict costs</i>
Purely transitory inflation	−0.73%	−0.10%	−0.63%
Persistent inflation	−2.46%	−0.33%	−2.12%
2021-23 inflation surge	−5.85%	−0.79%	−5.05%

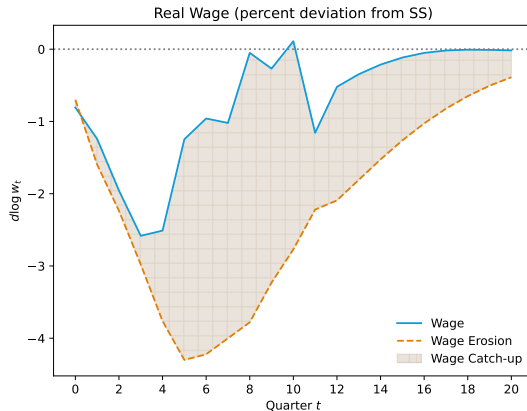
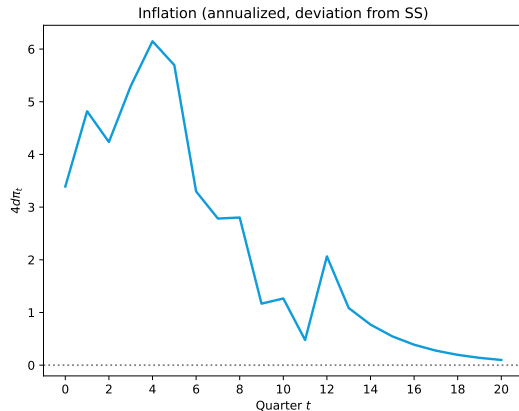
Wage Dynamics: Transitory Inflation with Sticky Expectations ($\lambda = 0.8$)

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Wage Dynamics: Persistent Inflation with Sticky Expectations ($\lambda = 0.8$)

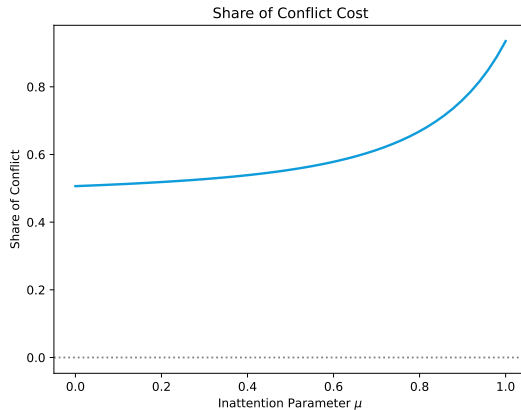
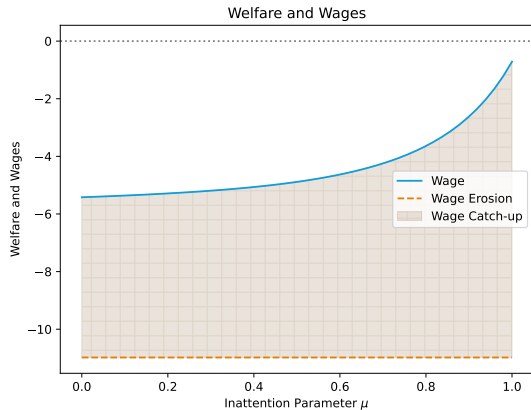
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Wage Dynamics: 2021-23 Inflation with Sticky Expectations ($\lambda = 0.8$)

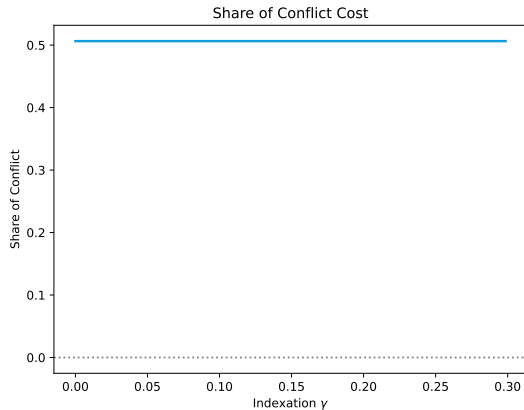
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	<i>Overall</i>	<i>Wage response</i>	<i>Conflict costs</i>
Purely transitory inflation	−0.98%	−0.06%	−0.92%
Persistent inflation	−3.30%	−0.71%	−2.60%
2021-23 inflation surge	−10.98%	−3.64%	−7.34%

Welfare losses during 2021-23 inflation surge as a function of inattention

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Welfare losses during 2021-23 inflation surge as a function of indexation

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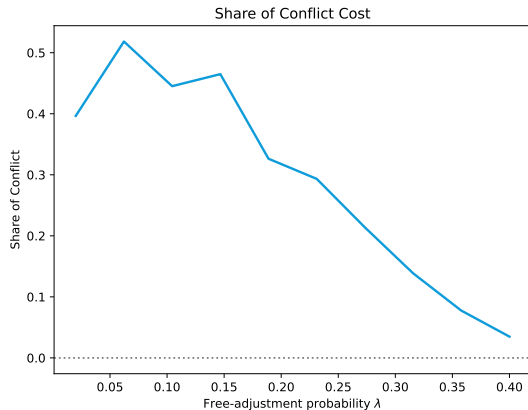
Proposition 2.

The ratio of aggregate costs of inflation due to conflict to its overall costs, $\hat{\pi}/\hat{\mathcal{W}}$, is invariant to the degree of indexation of default wage $\gamma \in [0, 1)$.

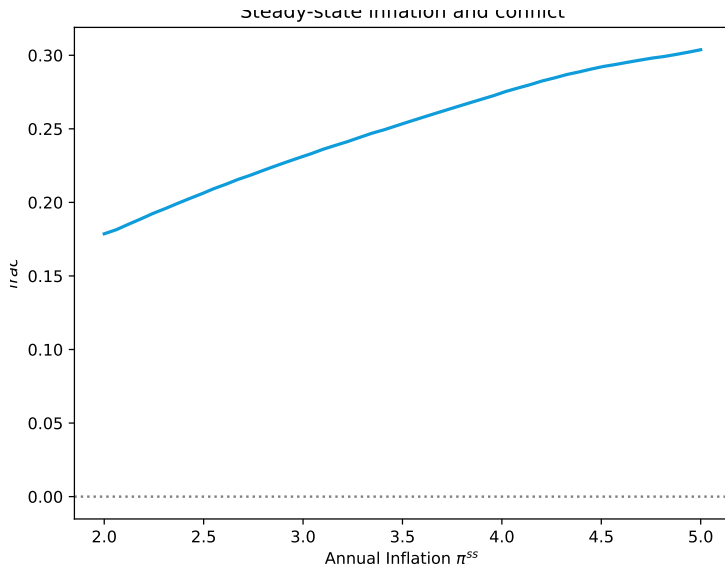
Intuition:

- All that matters for workers is *inflation net-of-indexation* $\hat{\pi}_t^{net} \equiv (1 - \gamma) \hat{\pi}_t$. Indexation scales down the “effective shock”.
- So, both overall costs of inflation and aggregate costs due to conflict scale proportionally with $1 - \gamma$.
- It follows that the ratio is invariant to γ .

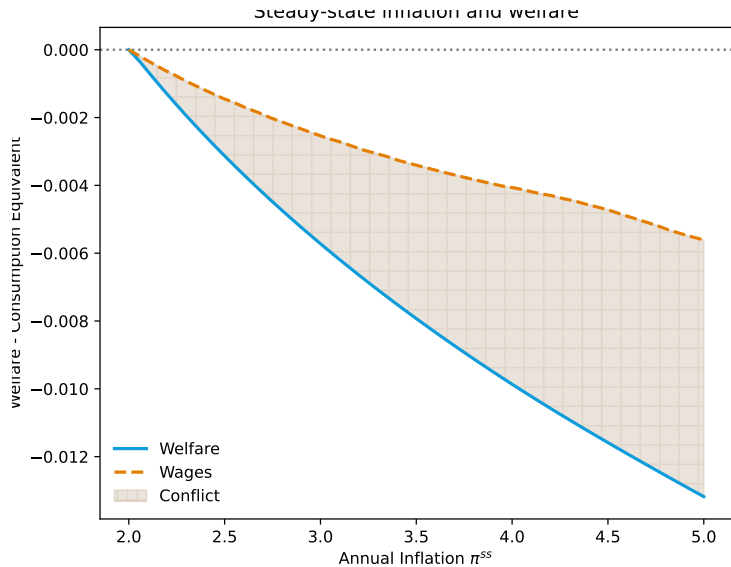
Welfare losses during 2021-23 inflation surge as a function of the free adjustment prob.

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Steady State Inflation and Probability of Conflict



Steady State Inflation and Costs of Conflict



- Service Employees International Union (≈ 1.9 million members): 1.7% of the member's gross monthly salary for time in regular pay status.
- International Brotherhood of Teamsters (≈ 1.4 million members): if you make \$15 an hour, your dues are $\approx 1.5\%$ of income.
- United Auto Workers (≈ 990 k members): two hours pay per month for full time members and 1.15% of straight-time hourly pay for part time and per diem members.
- National Education Association (≈ 3 million members): National dues are \$15 per year. For Chicago, to be a member costs from \$745.20 to \$1,242.00 a year.

Conflict Costs of Inflation in General Equilibrium

Conflict Costs in General Equilibrium

Goal of the GE analysis:

- Aggregate welfare
- “Inflation greases the wheels of the labor market” – allow employment to increase with inflation
- Endogenize conflict-induced wage

Overview of GE model:

- Same worker problem for employed, augmented with search & matching for unemployed
- Firms post vacancies and hire workers
- Agg. shocks to monetary policy & firm productivity, summarized by $\{\pi_t, y_t\}_{t=0}^{+\infty}$

Worker problem

Same as in PE model, adjusted with possibility of unemployment.

$$w_{i,t} = \begin{cases} w_{i,t-1} e^{\alpha - (1-\gamma)\pi_t} & \text{if } e_{i,t} = 1 \text{ and } \mathcal{I}_{i,t} = 0 \\ w_{i,t}^* & \text{if } e_{i,t} = 1 \text{ and } \mathcal{I}_{i,t} = 1 \\ \phi w_{i,t}^* & \text{if } e_{i,t} = 0 \end{cases}$$

Timing:

1. Employed workers separate with prob. s
2. Old and newly unemployed find jobs with prob. $f_t = f(\sigma_t)$ ($\sigma_t \sim$ market tightness)
3. Old workers choose to conflict with cost $\kappa_{i,t}$; new hires get conflict-induced wage ($\kappa_{i,t} = 0$)
4. Production takes place, wages are paid, and workers consume

Firms, Wages, and Employment

- Each firm is a match

$$J_t(w_{i,t}, \theta_{i,t}) = y_t \theta_{i,t} - w_{i,t} + \beta(1-s) \mathbb{E} \left[\mathcal{J}_{i,t+1} J_{t+1}(w_{i,t+1}^*, \theta_{i,t+1}) + (1 - \mathcal{J}_{i,t+1}) J_{t+1}(w_{i,t} e^{\alpha - (1-\gamma)\pi_{t+1}}, \theta_{i,t+1}) \right]$$

- Free entry condition

$$\mathbb{E} \left[J_t(w_{i,t}^*, \theta_{i,t}) \right] = \frac{c_v}{q_t}$$

where $q_t \sim$ job-filling rate

- Wage rule as in Dupraz et al. (2021)

$$\log \left(\frac{w_t^*}{w^{*,ss}} \right) = (1 - \rho) \log \left(\frac{y_t}{y^{ss}} \right) + \rho \log \left(\frac{w_{t-1}^*}{w^{*,ss}} \right)$$

and $w_{i,t}^* = w_t^* \theta_{i,t}$.

- Total employment

$$E_t = [1 - s(1 - f_t)] E_{t-1} + f_t [1 - E_{t-1}].$$

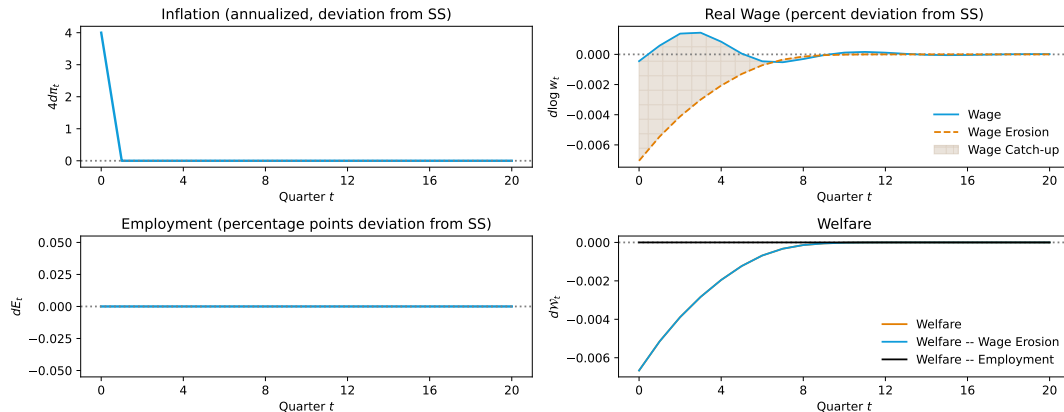
Worker Welfare

$$\mathcal{W} = \sum_{t=0}^{\infty} \beta^t \left[E_t \cdot \log w_t - \int_0^1 \kappa_{i,t} \mathcal{J}_{i,t} di + (1 - E_t) \cdot \log(\phi w_t^*) \right]$$

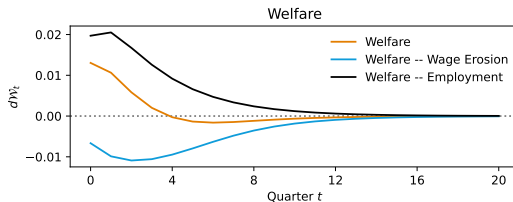
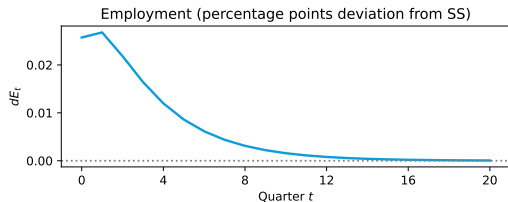
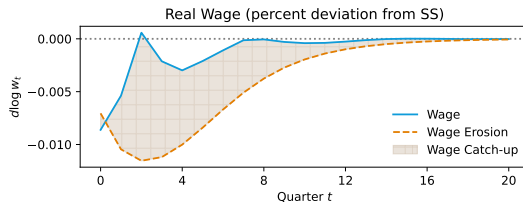
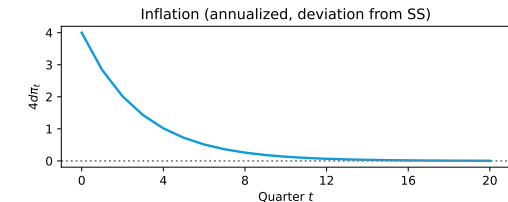
where $\log w_t = \int_0^1 e_{i,t} \log w_{i,t} di / E_t$, so

$$\begin{aligned} d\mathcal{W} &= \sum_{t=0}^{\infty} \beta^t \left[E_t \cdot \hat{w}_t^{\text{erosion}} - \log(\phi) \cdot dE_t \right] \\ &= \sum_{t=0}^{\infty} \beta^t \left[\underbrace{E_t \cdot (\hat{w}_t - \hat{w}_t^{\text{catch up}})}_{\text{wage erosion for employed}} - \underbrace{\log(\phi) \cdot dE_t}_{\text{how inflation affects employment rate}} \right] \end{aligned}$$

Worker Welfare in General Equilibrium: One-period Inflation



Worker Welfare in General Equilibrium: Persistent Inflation



Worker Welfare in General Equilibrium: 2021-23 Inflation

